JACK ERSBO
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PROFESSIONAL SUMMARY

I'm a detail-oriented and innovative software developer with strong foundations in programming languages and modern development tools. Currently, I'm supporting complex tax and finance systems for the Wisconsin Department of Revenue, with a track record of delivering high-impact features and enhancements on time and to specification. I'm experienced in full-stack development, application pipeline management, and collaborating with both technical and non-technical stakeholders to translate business needs into scalable software solutions. My background in academic research further developed advanced problem-solving, data analysis, and technical communication skills. I'm passionate about building efficient, secure, and user-focused applications that solve real-world problems.

EDUCATION

UNIVERSITY OF WISCONSIN, COLLEGE OF ENGINEERING — Madison, WI
Web Development Certificate — May 2022
Courses: HTML5, CSS3, JavaScript/TypeScript, Python, React.js., Redux, APIs, UI/UX Design

UNIVERSITY OF WISCONSIN-MADISON — Madison, WI
Bachelor of Science, Neurobiology — May 2020
Certificate, Global Health Studies

SKILLS PROFILE

Languages: Java, JavaScript/TypeScript, Python
Databases: Oracle SQL, PL/SQL scripting, PostgreSQL, MySQL, SQLite, Microsoft SQLServer
DevOps: Maven, Jenkins, Github Actions, Docker, AWS
Project management: Jira

PROFESSIONAL EXPERIENCE

Wisconsin Dept. of Revenue, Madison, WI — February 2023 - Present
Software Developer (IS Systems Dev Senior)
Developer on key tax and finance application projects, contributing to the full IT project lifecycle—from requirements gathering and architecture/design through development, testing, deployment, and ongoing support.

- Delivered features across the technology stack while adhering to compliance and security standards. Met deliverable and timeline requirements 100% of the time.
- Collaborated with stakeholders to gather technical and business requirements, propose architectural solutions, and deliver functional prototypes.
- Designed and implemented multiple first-of-kind enhancements to State and Local Finance Bureau systems, including implementing the ability for Wisconsin tribal governments to apply for state aid programs, and designing and implementing new security standards to facilitate the storage of personal identifying information.
- Participated in technical planning and helped manage application pipelines.
- Provided informal team leadership by mentoring peers, reviewing code, and helping communicate technical considerations to non-technical stakeholders.

Univ. Wisconsin Laryngeal Physiology Lab, Madison, WI May 2020 - July 2022

Lab Manager and Research Specialist

- Fast-paced, engineering-driven, collaborative environment.
- Helped maintain the lab's codebases (Python, Matlab, LabView, and C#). Built and deployed small Node.js web applications to help with human subject recruitment and data collection. Took initiative to implement new features in the lab's staff website and created documentation for future managers.
- Managed up to 25 undergraduate researchers; tracked their progress and gave direction when they needed help. Hiring manager for students and full-time staff.
- Key person in grants administration consisting of reports to the NIH, PHS, and other funding agencies (3 RO1 grants totaling over $4 million in total support).
- Oversaw the production and publication of over a dozen scientific articles. Helped coordinate academic collaborations with other universities, both domestic and foreign.
- Published my own work—an example below:
  Fang, Yanqing; Zhang, Kun; **Ersbo, Jack H.**; Chen, Bing. *The Impact of the Frequency-Specific Preoperative Sensorineural Hearing Loss to Postoperative Overclosure of Bone Conduction in Stapedotomy*. Otology & Neurotology: October 2021 - Volume 42 - Issue 9 - p 1314-1322

Univ. Wisconsin Laryngeal Physiology Lab, Madison, WI April 2018 – May 2020

Undergraduate Research Assistant (Student Engineer, Programmer)

- Pursued both independent and collaborative research. Led one of the lab's research teams.
- Specialized in instrumentation and imaging. Adapted new techniques for studying laryngeal tissue composition including XCLARITY, MYOCLEAR, OrgoClear (proprietary), Optical Coherence Tomography, Tissue Dielectric Property measurement.
- Created data collection and analysis programs which interfaced with experimental hardware.
- Published one manuscript and gave two conference poster presentations.